PMP CERTIFICATION COURSE

Student Reference Resource Guide

Section 3:
Life Cycles, Focus Areas & Development Approaches

All Lectures Covered

PMBOK Guide 8th Edition • 2026 Examination Content Outline

Prepared for PMP Exam Candidates

Section 3 Overview

Section 3 of the PMP Certification Course addresses one of the most practically impactful areas of project management: how to structure and approach the work of a project from start to finish. The section covers project life cycles (phases and phase gates), development approaches (predictive, adaptive, and hybrid), delivery cadence, and the five focus areas (Initiating, Planning, Executing, Monitoring & Controlling, and Closing) — examining each individually and as an integrated system.

Why does life cycle choice matter? The development approach you select determines how your project is planned, how work is executed, how change is managed, and when value is delivered to stakeholders. Choosing predictive when requirements are volatile leads to scope creep and rework. Choosing adaptive when regulatory traceability is required creates compliance risk. The ability to assess project context and select — or tailor — the right approach is a core competency tested on the PMP exam.

Connection to Tailoring (PMBOK 8 Principle 5): PMBOK Guide 8th Edition places tailoring at the heart of modern project management. No project is the same, and no single methodology fits every situation. Section 3 provides the vocabulary and framework for tailoring decisions — understanding phases, gates, approaches, and focus areas enables project managers to make informed, context-sensitive choices rather than defaulting to organizational habit.

How Section 3 shows up on the PMP exam: Expect 15-25% of PMP questions to involve development approach selection, life cycle management, or focus area activities. Questions are scenario-based and application-focused — they test whether you can choose the right approach for a given situation, not just define terms. Pay special attention to distinguishing predictive vs. adaptive behaviors, EVM calculations in Monitoring & Controlling, and the differences between agile ceremonies and traditional project management activities.

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Lecture 1: Project Phases and Phase Gates

Overview

In the world of project management, understanding how a project is divided into phases and how those phases are controlled is fundamental to success. A project phase is a distinct sequence of activities that produces one or more deliverables. Phases provide structure, making it easier to manage complex projects by breaking them into manageable segments with defined objectives, resources, and timelines.

Phase gates — sometimes called stage gates, kill points, or phase reviews — are the formal checkpoints between phases. At each gate, the project sponsor, steering committee, or governance body evaluates the deliverables produced in the completed phase and decides whether to authorize the next phase. This ensures that investment continues only when the project remains aligned with organizational strategy and business value.

The way phases and gates are structured is closely linked to your chosen development approach (predictive, adaptive, or hybrid). In a predictive approach, phases tend to be sequential with formal gates. In an adaptive approach, phases may overlap, occur concurrently, or be replaced by iterative sprint cycles. The PMP exam will test your ability to select the right phase structure based on project context — making this concept foundational for Section 3.

Key Concepts

- **Project Phase Definition:** A distinct, sequential group of project activities with defined start/end points and measurable deliverables. Phases divide the project lifecycle chronologically.
- **Phase Characteristics:** Phases are time-bound, resource-intensive, and produce specific deliverables. They may be sequential (predictive) or overlapping (adaptive/hybrid).
- **Phase Gate (Stage Gate):** A formal review point at the end of each phase where project performance is evaluated against criteria. Decisions: Go (proceed), No-Go (stop), Hold (pause), or Recycle (redo phase).
- **Phase-End Reviews:** Formal assessments using deliverables and performance data to determine project health before committing resources to the next phase.
- **Sequential vs. Overlapping Phases:** Predictive projects run phases sequentially with handoffs; adaptive projects may overlap or run phases in parallel to accelerate delivery.
- **Tailoring Phase Structure:** Phase structures are tailored to the project context — industry norms, organizational standards, regulatory requirements, risk tolerance, and complexity all influence phase design.
- **Deliverables as Phase Milestones:** Each phase produces measurable outputs (documents, prototypes, completed work) that serve as evidence of phase completion and basis for gate reviews.
- **Project vs. Product Life Cycle:** The project life cycle covers the duration of the project itself; the product life cycle spans from concept through retirement. Understanding both is critical to exam success.
- **Milestone vs. Phase Gate:** A milestone marks a significant point in the schedule; a phase gate is a go/no-go decision checkpoint. All phase gates are milestones, but not all milestones are phase gates.

- **Governance at Phase Gates:** Senior stakeholders or governance boards (e.g., PMO) review phase gate criteria including scope, schedule, budget, risk, and strategic alignment.

Key Terms & Definitions

Term	Definition
Project Phase	A collection of logically related project activities culminating in the completion of one or more deliverables.
Phase Gate	A review at the end of a phase where project performance is evaluated before proceeding to the next phase. Also called stage gate, kill point, or phase review.
Go/No-Go Decision	The output of a phase gate review: authorize continuation, halt the project, put it on hold, or request rework.
Sequential Phases	Phases arranged so that one must fully complete before the next begins; typical of predictive (waterfall) approaches.
Overlapping Phases	Phases that run concurrently or begin before the prior phase ends; typical of adaptive or fast-track approaches.
Project Life Cycle	The series of phases from project initiation through project closure, regardless of development approach used.
Product Life Cycle	The full lifespan of a product from initial conception through market introduction, growth, maturity, and eventual retirement.
Phase-End Review	A structured evaluation at the end of each project phase to assess performance, validate deliverables, and make continuation decisions.
Governance Board	A group of senior stakeholders who provide oversight, strategic direction, and approval decisions at phase gates.

Exam Tips

✦ EXAM TIPS

- Phase gates and project phases are NOT the same. A phase is a body of work; a phase gate is the decision checkpoint between phases.
- On the exam, when asked about approving continuation of a project, the correct tool is a phase gate review — not a status meeting or change request.
- Predictive projects use formal, sequential phase gates. Adaptive projects use iteration reviews (sprint reviews) as their equivalent governance checkpoints.
- The PMP exam frequently tests the difference between the project life cycle and the product life cycle. The project ends at closeout; the product continues.

- Remember that tailoring is always the answer when the exam asks why two organizations or projects have different phase structures.
- Phase gates reduce risk by ensuring resources are only committed to work that still delivers value — connect this concept to stakeholder value (Principle 1 of PMBOK 8).

Knowledge Check

✦ KNOWLEDGE CHECK

1. A project sponsor asks you to proceed directly to the construction phase of a building project without completing the design phase review. What is the PRIMARY concern with this approach, and what should you do?
2. Your agile software project uses two-week sprints with sprint reviews. A traditional PMO asks why you don't have formal phase gates. How would you best explain the relationship between sprint reviews and phase gates?
3. During a phase gate review, the governance board identifies that the project's business case has changed and ROI has dropped significantly. What are the possible outcomes of this review?
4. A project manager is designing the phase structure for a new pharmaceutical product. What organizational factors should most influence how many phases and phase gates are included?
5. A project uses overlapping phases to compress the schedule. What is the PRIMARY risk of this approach, and how should it be managed?

Lecture 2: Development Approaches — Predictive, Adaptive and Hybrid

Overview

The development approach is one of the most strategically important decisions a project manager makes. It defines how a project will be planned, executed, and controlled throughout its lifecycle. PMBOK Guide 8th Edition recognizes three primary approaches: predictive, adaptive, and hybrid — each suited to different levels of uncertainty, stakeholder engagement, and delivery needs.

The predictive approach (also called waterfall or plan-driven) works best when requirements are well-understood upfront, risk is manageable, and sequential delivery is acceptable. The adaptive approach (agile) thrives in high-uncertainty, rapidly changing environments where frequent delivery of working increments provides faster value and enables learning. The hybrid approach blends elements of both, allowing teams to match their methodology to the unique characteristics of different project components.

On the PMP exam, choosing or recommending the right development approach is a frequently tested scenario. You must understand not just the definitions, but the rationale behind selecting each approach based on project, organizational, and environmental factors. This lecture establishes the conceptual foundation on which the rest of Section 3 is built.

Key Concepts

- **Predictive Approach (Waterfall):** A sequential development approach where requirements are defined upfront, plans are created before execution, and deliverables are produced at the end of the project. Best for stable, well-understood requirements.
- **Adaptive Approach (Agile):** An iterative/incremental approach that delivers working features in short cycles (sprints/iterations). Requirements evolve through collaboration. Best for high-uncertainty, innovative, or rapidly changing environments.
- **Hybrid Approach:** A combination of predictive and adaptive practices tailored to project needs. Example: predictive for procurement/compliance work, agile for development/design.
- **Iterative vs. Incremental:** Iterative development refines work through repeated cycles; incremental development delivers work in separate functional pieces. Agile is both iterative AND incremental.
- **Fixed vs. Flexible Scope:** Predictive fixes scope early; adaptive keeps scope flexible to respond to change. Budget and schedule flexibility differ accordingly.
- **Velocity and Throughput:** Agile teams measure capacity using velocity (story points completed per sprint) or throughput (number of items delivered). These metrics guide planning in adaptive projects.
- **Iron Triangle Differences:** In predictive: scope is fixed, time and cost may vary. In adaptive: time and cost are fixed per iteration, scope varies. This is critical for exam scenarios.
- **Servant Leadership in Adaptive Projects:** Agile project managers act as servant leaders and coaches rather than command-and-control managers. The team self-organizes within the sprint framework.

- **Stakeholder Engagement Frequency:** Adaptive approaches require frequent stakeholder engagement (every sprint). Predictive may have formal review points only at phase gates.
- **Change Management:** Predictive uses formal change control boards (CCBs) for scope changes. Adaptive welcomes change via backlog refinement — even late in development.

Key Terms & Definitions

Term	Definition
Predictive Approach	A development approach that fully defines scope, schedule, and cost upfront. Changes are controlled through a formal change control process.
Adaptive Approach	An agile, iterative development approach that delivers incremental value and responds to change through regular stakeholder feedback.
Hybrid Approach	A tailored blend of predictive and adaptive practices applied to different parts of the same project based on their specific characteristics.
Sprint / Iteration	A fixed-length development cycle (typically 1-4 weeks) in adaptive projects during which a potentially releasable increment of the product is developed.
Backlog	An ordered list of all work to be done on the project, maintained and prioritized by the Product Owner in adaptive approaches.
Velocity	A measure of the amount of work an agile team can complete in a single sprint, used for capacity planning.
Increment	A functional, potentially releasable piece of the product delivered at the end of each iteration in adaptive development.
Iron Triangle	The constraint relationship between scope, time, and cost. Scope is fixed in predictive; time and cost per iteration are fixed in adaptive.
Change Control Board (CCB)	A group responsible for reviewing and approving/rejecting change requests in predictive project environments.

Exam Tips

✦ EXAM TIPS

- The PMP exam heavily tests when to use each approach. If the scenario mentions changing requirements, high uncertainty, or customer collaboration — lean toward adaptive.
- If the scenario mentions regulatory compliance, fixed budget, stable requirements, or physical construction — lean toward predictive.

- Hybrid is correct when different parts of the same project have different uncertainty levels (e.g., agile for software development + predictive for hardware procurement).
- Remember the Iron Triangle flip: predictive fixes scope; adaptive fixes time/cost per sprint but allows scope to evolve.
- Never say agile means 'no planning' on the exam. Adaptive approaches still plan — at multiple levels (release plan, sprint plan, daily standup).
- The exam may ask about servant leadership specifically in agile contexts. The PM coaches the team rather than directing work.

Knowledge Check

✦ KNOWLEDGE CHECK

1. A client insists on approving all requirements before development begins, but also expects the product to adapt to market changes during the 18-month project. Which development approach would you recommend, and why?
2. A government infrastructure project has fixed regulatory requirements, a fixed budget approved by legislature, and a 3-year timeline. Which development approach is most appropriate?
3. Your organization runs two parallel workstreams: hardware procurement (fixed specs) and software development (evolving requirements). How would you structure the development approach?
4. An agile team's velocity has been consistently 40 story points per sprint. The product backlog contains 320 story points. What is the team's projected delivery timeline if sprints are 2 weeks long?
5. A project manager transitioning from predictive to adaptive is struggling with the concept of 'fixed scope.' How would you explain the adaptive approach's treatment of scope to this PM?

Lecture 3: Considerations for Development Approach Selection

Overview

Selecting the right development approach is not a matter of personal preference — it is a structured decision that requires evaluation of multiple factors spanning the project, product, team, and organization. PMBOK Guide 8th Edition provides a framework of considerations that guide this critical decision, ensuring alignment between how work is done and the realities of the project environment.

The key consideration areas include: the nature of the product or service being delivered, the complexity and uncertainty inherent in the work, the capabilities and preferences of the team, the organizational culture and governance structure, and the needs and engagement preferences of stakeholders. Each factor can push a project toward predictive, adaptive, or hybrid approaches.

On the PMP exam, development approach selection scenarios are frequent and nuanced. The exam rarely presents clear-cut cases — instead, it tests your ability to weigh competing factors and make the best recommendation given constraints. Understanding the full range of selection factors will enable you to justify your approach choice, tailor appropriately, and explain that decision to stakeholders.

Key Concepts

- **Product Factors:** The nature of the product significantly influences approach: products requiring physical integration (construction, manufacturing) favor predictive; software or innovation-heavy products favor adaptive.
- **Uncertainty and Risk:** Higher uncertainty in requirements or technology favors adaptive (frequent feedback reduces risk). Lower uncertainty with proven technology favors predictive.
- **Team Capability:** Agile approaches require team members who are self-organizing, cross-functional, and comfortable with ambiguity. Predictive suits teams with specialized, sequential roles.
- **Organizational Culture:** Command-and-control cultures favor predictive; collaborative, empowerment-driven cultures favor adaptive. Culture change may be needed before adoption.
- **Stakeholder Engagement:** Stakeholders who can be actively involved sprint-by-sprint favor adaptive. Stakeholders who prefer defined deliverables at specified dates favor predictive.
- **Regulatory and Compliance Environment:** High-regulation industries (pharmaceuticals, aviation, defense) often require predictive approaches due to documentation and traceability requirements.
- **Team Size and Distribution:** Co-located, small teams can adopt full agile. Large, distributed, or multi-vendor teams may need more predictive structure and formal communication plans.
- **Delivery Timing Requirements:** If the customer needs all deliverables at once, predictive works. If incremental value delivery is acceptable or preferred, adaptive is advantageous.

- **Organizational Maturity:** Organizations new to agile may need a hybrid bridge. Mature agile organizations can adopt fully adaptive approaches across the portfolio.
- **Tailoring Principle:** No single approach fits all projects. PMBOK Principle 5 (Tailoring) requires actively adapting the approach based on these selection factors — not defaulting to what the team is comfortable with.

Key Terms & Definitions

Term	Definition
Development Approach Selection	The process of choosing predictive, adaptive, or hybrid based on project, product, team, and organizational factors.
Tailoring	Intentional adaptation of project management processes, methods, and approaches to the specific context of the project.
Organizational Culture	The shared values, beliefs, and behaviors of an organization that influence how work is done and how teams prefer to be managed.
Regulatory Compliance	External legal or industry requirements that may constrain development approach choice, typically favoring predictive for traceability.
Team Maturity	The collective capability, experience, and self-organization of a project team; high maturity supports adaptive approaches.
Stakeholder Engagement Model	The approach to how and how frequently stakeholders are involved in project decisions and reviews.
Delivery Timing	Whether the customer requires all deliverables at once (predictive) or can benefit from incremental releases (adaptive).
Complexity	The degree of interconnectedness, uncertainty, and dynamic behavior in a project; high complexity favors adaptive or iterative approaches.
Product Novelty	The degree to which the product is new or unprecedented; higher novelty favors adaptive to allow for learning and adjustment.

Exam Tips

✦ EXAM TIPS

- When the exam describes a project with 'unclear requirements' or 'rapidly changing market conditions,' the answer is almost always adaptive or hybrid — not predictive.
- Watch for regulatory keywords: 'FDA approval,' 'FAA certification,' 'government audit trail' — these signal predictive even if other factors suggest adaptive.
- Team capability is an often-overlooked factor. If the team is 'new to agile' or 'distributed across time zones,' a hybrid or predictive approach may be more practical.

- Never select an approach purely on organizational preference. The correct answer always balances multiple factors — the exam penalizes single-factor reasoning.
- Remember: Tailoring is not optional. PMBOK 8 explicitly positions tailoring as a core activity, not an exception.
- A hybrid approach is always valid as a compromise when pressures from different directions (regulatory + innovation) conflict.

Knowledge Check

✦ KNOWLEDGE CHECK

1. A pharma company is developing a new drug delivery device. Software components have evolving requirements; the physical device must meet FDA regulatory standards. What development approach framework would you recommend?
2. A team of 5 co-located developers with 3 years of agile experience is building a mobile app for a startup. Customer requirements change weekly. Which approach and why?
3. A project manager proposes using predictive because 'that is what our organization has always done.' What is wrong with this reasoning according to PMBOK 8?
4. You are evaluating two projects: Project A has stable requirements, a government contract, and a fixed delivery date. Project B has innovative features, evolving user stories, and a flexible timeline. What approach would you recommend for each?
5. An organization's culture is hierarchical and resistant to change. You believe an adaptive approach would benefit the project. What should you consider before recommending it?

Lecture 4: Delivery Cadence — Single, Multiple and Periodic Delivery

Overview

Delivery cadence refers to the timing and frequency at which project deliverables are provided to the customer or end users. PMBOK Guide 8th Edition distinguishes between three patterns: single delivery, multiple deliveries, and periodic deliveries. Each pattern directly influences which development approach is most appropriate and how value is realized throughout the project lifecycle.

Single delivery means the customer receives all project outcomes at the end of the project — typical of construction, manufacturing, or large infrastructure projects. Multiple delivery involves releasing different components or phases at different points in time, often used in product rollouts or phased deployments. Periodic delivery, characteristic of agile approaches, provides regular increments of value at fixed intervals (such as every two weeks), enabling continuous feedback and course correction.

Understanding delivery cadence is critical for the PMP exam because it links directly to development approach selection, stakeholder value delivery, and the concept of benefits realization. The exam will test your ability to recommend or recognize the appropriate cadence for a given project scenario, and to connect cadence decisions to the broader strategic goal of delivering maximum value.

Key Concepts

- **Single Delivery:** All deliverables are provided to the customer as one complete package at the end of the project. Typical of physical construction, infrastructure, and compliance-driven projects.
- **Multiple Delivery:** Project deliverables are released in discrete batches at different points in time. Allows staged implementation or phased rollout across regions or business units.
- **Periodic Delivery (Agile Cadence):** Regular, fixed-interval release of working product increments. Typical of agile/scrum approaches with 1-4 week sprints. Enables rapid feedback loops.
- **Value Realization Timing:** Delivery cadence directly determines when the customer begins realizing value. Periodic delivery allows earlier and more continuous value realization than single delivery.
- **Feedback Loops and Learning:** Periodic delivery creates structured feedback loops — each increment provides an opportunity for the customer to validate direction and suggest course corrections.
- **Cadence and Approach Alignment:** Single delivery aligns with predictive; periodic delivery aligns with adaptive; multiple delivery is common in hybrid approaches.
- **Release vs. Iteration:** In adaptive projects, an iteration is an internal development cycle (sprint); a release is when working software is delivered to the customer. Releases may span multiple iterations.
- **Cadence and Risk:** Periodic delivery reduces risk by limiting the amount of work in progress and enabling early course correction. Single delivery concentrates risk at the end of the project.

- **Customer Capability to Absorb Delivery:** Some customers cannot absorb frequent deliveries due to organizational capacity, training needs, or integration constraints — this must be factored into cadence selection.
- **Continuous Deployment (CD):** An advanced form of periodic delivery where working software is released continuously (sometimes multiple times per day) through automated pipelines.

Key Terms & Definitions

Term	Definition
Delivery Cadence	The timing and frequency with which project deliverables are provided to the customer or end user.
Single Delivery	A delivery pattern where all project deliverables are provided once at the end of the project lifecycle.
Multiple Delivery	A delivery pattern where deliverables are released in distinct batches at multiple points during the project.
Periodic Delivery	A regular, fixed-interval delivery pattern in which working increments are provided at the end of each iteration or sprint.
Release	A formal delivery of a product increment to the customer or end users; may encompass one or more completed iterations.
Iteration	A fixed-length development cycle (sprint) in adaptive projects; the internal unit of work before customer-facing release.
Value Realization	The point at which customers or the organization begin benefiting from the project's deliverables.
Feedback Loop	The mechanism by which customer or stakeholder input on delivered increments informs future development priorities.
Continuous Deployment	An automated delivery pattern where every approved code change is immediately deployed to production.

Exam Tips

✦ EXAM TIPS

- The exam will present delivery cadence questions as scenario-based choices. If the question mentions 'value as soon as possible' or 'early feedback,' choose periodic delivery and adaptive approach.
- Single delivery is NOT always wrong — for a bridge, dam, or regulatory filing, single delivery is the only logical option. Context is everything.
- Know the distinction between a release and a sprint/iteration. Teams may sprint every 2 weeks but release to customers monthly — both are valid cadence choices.

- Periodic delivery reduces risk because defects and misalignments are caught earlier. This connects to the exam's broader theme of proactive risk management.
- If the exam asks about 'customer satisfaction' and the scenario allows it, periodic delivery is almost always the preferred cadence because it enables feedback and adaptation.
- Watch for customer absorption capacity — a technically correct answer (periodic delivery) can be wrong if the customer 'cannot handle frequent changes' as stated in the scenario.

Knowledge Check

✦ KNOWLEDGE CHECK

1. A customer asks for a project that 'delivers something useful every month.' What delivery cadence does this suggest, and which development approach best supports it?
2. A highway construction project will deliver the fully completed road in 30 months. The client wants no interim deliverables. What delivery cadence applies, and what are the risk implications?
3. A software team releases to production every two weeks (sprints) but customer-facing releases happen quarterly due to their internal change management process. How many delivery cadences are at play here?
4. A project manager argues that single delivery is safer because 'we can perfect everything before handing it over.' How would you counter this argument using delivery cadence theory?
5. A retail company wants to roll out a new inventory system to 5 regional warehouses over 6 months — one warehouse per month. What delivery pattern does this represent, and what approach would you use?

Lecture 5: Focus Area 1 — Initiating

Overview

The Initiating focus area represents the set of activities that formally authorize a project or phase and set the direction for how work will proceed. In PMBOK Guide 8th Edition, focus areas replace the traditional process groups but cover the same foundational activities — they describe WHAT needs to be done rather than prescribing HOW or WHEN. Initiating activities ensure that the project has clear purpose, defined stakeholders, assigned leadership, and organizational commitment before resources are invested.

Key activities in the Initiating focus area include developing the Project Charter, identifying and analyzing stakeholders, establishing the preliminary scope, defining high-level risks, and forming the initial project team. These activities are not restricted to the start of the project — in adaptive projects, initiating activities recur at the start of each major release or phase. In predictive projects, initiating occurs once at the project level and optionally at each phase.

The Initiating focus area is particularly important on the PMP exam because it establishes the authority of the project manager, the organizational commitment to the project, and the strategic link between the project and business objectives. Many exam questions center on what happens when initiating is skipped, done poorly, or done without proper stakeholder involvement — all of which lead to project failure patterns.

Key Concepts

- **Project Charter:** The document that formally authorizes the project, grants the project manager authority, and links the project to organizational strategy. Created during initiating.
- **Stakeholder Identification:** The process of discovering all individuals, groups, or organizations affected by the project. Initial identification happens during initiating; continues throughout the project.
- **Preliminary Scope:** A high-level description of what the project will and will not deliver, developed during initiating before detailed planning begins.
- **Business Case:** The document that justifies the project investment by identifying the problem, opportunity, and expected benefits. Precedes the charter.
- **Benefits Realization Plan:** A plan describing how and when project benefits will be realized. Initiating establishes initial benefit expectations.
- **Project Manager Authority:** Defined in the Project Charter — the PM has the authority they are explicitly granted. Without a charter, the PM lacks formal organizational authority.
- **High-Level Risk Identification:** During initiating, major risks (threats and opportunities) are identified at a high level. Detailed risk analysis occurs in planning.
- **Assumptions and Constraints:** Initial assumptions (what is believed to be true without evidence) and constraints (limitations on the project) are documented during initiating.
- **Project Statement of Work (SOW):** A description of products, services, or results to be supplied by the project. Often an input to the charter.
- **Lessons Learned from Prior Projects:** Historical information from previous projects informs the initiating process, helping avoid known pitfalls and leverage proven approaches.

Key Terms & Definitions

Term	Definition
Project Charter	A formal document that authorizes the existence of a project, names the project manager, and provides high-level scope, objectives, and authority levels.
Stakeholder	Any individual, group, or organization that may affect, be affected by, or perceive itself to be affected by a project decision or outcome.
Business Case	A documented economic feasibility study that captures the reasons for undertaking the project and the expected business benefits.
Initiating	The focus area comprising activities to define a new project or phase, obtain authorization, and identify initial stakeholders and requirements.
Assumptions Log	A project document that records all project assumptions and constraints identified throughout the project lifecycle, starting in initiating.
Benefits Realization	The process of ensuring that project outputs translate into intended business value and outcomes.
Project Sponsor	The executive who provides resources and support for the project and is accountable for enabling success at the organizational level.
Feasibility Study	An analysis conducted before project initiation to determine whether the project is viable technically, financially, and organizationally.
SOW (Statement of Work)	A narrative description of the products, services, or results to be delivered by a project.

Exam Tips

✦ EXAM TIPS

- The Project Charter is THE most important output of initiating. The PM should always verify it exists and is signed by the sponsor before starting any other work.
- If the exam asks who creates the Project Charter, it is NOT the project manager — it is the sponsor or initiating organization. The PM may assist in developing it.
- Stakeholder identification begins in initiating but NEVER stops. The exam may present late-identified stakeholders — always engage them regardless of when they appear.
- When exam questions describe a PM without authority, without organizational support, or with no clear project definition — this signals that initiating was poorly done.
- The business case precedes the charter. The charter precedes planning. Sequence matters on the PMP exam.

- A project without a charter (informal projects) still needs the PM to seek written authorization — this is always the correct answer when the question involves informal project starts.

Knowledge Check

✦ KNOWLEDGE CHECK

1. A project manager begins planning activities before a Project Charter has been signed. What is the PRIMARY risk of this approach?

2. You discover midway through project execution that a major government agency will be significantly impacted by your project, but was not identified during initiating. What should you do?

3. The project sponsor asks you to begin work immediately because 'the charter is just a formality.' How do you respond, and what are the consequences of proceeding without one?

4. A business case identifies three project options with different ROI profiles. Who is responsible for selecting which option to pursue, and what document formalizes the decision to proceed?

5. On a large program, each project within the program will have its own charter. What should these individual project charters reference, and how should they align to the program?

Lecture 6: Focus Area 2 — Planning

Overview

The Planning focus area encompasses the activities required to establish the total scope of effort, define and refine objectives, and develop the course of action required to achieve those objectives. Planning is the intellectual heart of project management — it transforms the high-level direction set during initiating into actionable plans that guide execution, monitoring, and control. In PMBOK Guide 8th Edition, planning is recognized as a recursive, iterative activity that continues throughout the project, not a one-time event.

In predictive projects, planning is front-loaded: comprehensive plans are developed early and guide execution for the duration of the project. In adaptive projects, planning occurs at multiple levels simultaneously — strategic/release planning at the beginning, sprint planning at the start of each iteration, and daily planning during standups. The key distinction is that adaptive planning embraces uncertainty by planning in detail only for near-term work while keeping long-term planning flexible.

The PMP exam tests planning from multiple angles: what should be planned and when, how plans are integrated across the knowledge areas, how baselines are established and protected, and how planning differs between predictive and adaptive approaches. Understanding planning as a dynamic, continuous activity rather than a static deliverable is the key mental shift candidates must make.

Key Concepts

- **Project Management Plan:** The master planning document that describes how the project will be executed, monitored, controlled, and closed. Integrates all subsidiary plans and baselines.
- **Scope Planning and WBS:** Defining the full scope of work and breaking it into the Work Breakdown Structure (WBS) — a hierarchical decomposition of all deliverables.
- **Schedule Planning:** Sequencing activities, estimating durations, and developing the project schedule. Tools include network diagrams, critical path method (CPM), and Gantt charts.
- **Cost Planning and Budgeting:** Estimating costs for each activity and work package, then aggregating to establish the cost baseline and overall project budget.
- **Risk Planning:** Identifying, analyzing (qualitatively and quantitatively), and planning responses to both threats and opportunities.
- **Quality Planning:** Defining quality standards, metrics, and control approaches to ensure deliverables meet requirements.
- **Resource Planning:** Identifying and acquiring the human and physical resources needed, defining roles and responsibilities.
- **Communications Planning:** Determining stakeholder communication needs and planning how, when, and through what channels information will be shared.
- **Rolling Wave Planning:** A planning technique where near-term work is planned in detail and future work is planned at a high level, with details added as more is known.
- **Baselines:** Approved versions of scope (scope baseline), schedule (schedule baseline), and cost (cost baseline). Changes to baselines require formal change control.

Key Terms & Definitions

Term	Definition
Project Management Plan	An integrated document approved by the project sponsor that defines how all aspects of the project will be managed, executed, and controlled.
Work Breakdown Structure (WBS)	A hierarchical decomposition of the total scope of work into manageable work packages, with WBS Dictionary providing details for each element.
Critical Path Method (CPM)	A schedule network analysis technique that identifies the longest path through the project network — delays on the critical path delay the project end date.
Cost Baseline	The approved, time-phased project budget against which actual costs are compared to identify variances.
Rolling Wave Planning	Progressive elaboration of the project plan where near-term work packages are planned in detail and future work is left at higher levels until more information is available.
Risk Register	A document tracking identified risks, their probability, impact, risk scores, response strategies, and risk owners.
Resource Management Plan	A component of the project management plan that describes how project resources will be acquired, allocated, managed, and released.
Change Control Plan	The section of the project management plan describing how changes to baselines will be submitted, evaluated, approved or rejected, and implemented.
Scope Baseline	The approved version of the scope statement, WBS, and WBS Dictionary against which scope changes are measured.

Exam Tips

✦ EXAM TIPS

- The PMP exam distinguishes between planning documents and planning activities. The Project Management Plan is not created once — it is progressively elaborated and updated throughout the project.
- Rolling wave planning is the PMBOK-endorsed bridge between predictive and adaptive. Expect exam questions where rolling wave is the correct planning technique in hybrid scenarios.
- Baselines (scope, schedule, cost) can only be changed through formal change control. Any change that shifts a baseline requires a change request — even if the sponsor verbally approves it.

- In adaptive planning, the 'plan' lives in the backlog. The sprint plan, release plan, and product roadmap together replace the monolithic project management plan.
- Exam questions about 'the PM skipped planning to start faster' almost always end badly — incomplete scope, missed risks, cost overruns. Planning is always worth the investment.
- Remember: planning outputs feed every other focus area. Poor planning is the root cause cited in most exam failure scenarios.

Knowledge Check

✦ KNOWLEDGE CHECK

1. A project manager is under pressure to begin execution immediately. The sponsor says 'planning takes too long.' How would you justify the time investment in planning to a skeptical sponsor?
2. A project using rolling wave planning has a detailed plan for the next 3 months but only high-level milestones for the remaining 9 months. A team member says this is 'incomplete.' How do you respond?
3. During planning, the project manager discovers that the project scope is significantly larger than what was described in the Project Charter. What should happen next?
4. What is the difference between the project management plan and the project schedule? When does each get created, and who approves each?
5. In an agile project, how does sprint planning differ from release planning? Which is more strategic and which is more tactical?

Lecture 7: Focus Area 3 — Executing

Overview

The Executing focus area covers the activities required to complete the work defined in the project management plan and deliver the project's products, services, or results. Execution is where the majority of the project budget is spent and where the vast majority of the project team's time is invested. PMBOK Guide 8th Edition recognizes that execution is not simply 'doing the work' — it involves active management, leadership, communication, issue resolution, quality assurance, and procurement.

In predictive projects, executing follows the approved plan with formal processes for managing changes, quality, and team performance. In adaptive projects, executing happens within sprints — the development team self-organizes to complete backlog items, facilitated by daily standups, sprint reviews, and retrospectives. The project manager's (or Scrum Master's) role in execution shifts from direction-giver to obstacle-remover.

The PMP exam places heavy emphasis on execution-related topics because this is where most project crises occur: scope creep, quality failures, team conflicts, vendor performance issues, and stakeholder dissatisfaction. Candidates must know how to recognize execution problems early and respond appropriately, using the right tools and escalation paths.

Key Concepts

- **Direct and Manage Project Work:** The primary execution process — leading and performing the work defined in the project management plan, managing change requests, and producing deliverables.
- **Quality Assurance (QA):** Process-oriented quality activity that audits quality processes and ensures the project is following the right methods to produce quality deliverables. Proactive, not reactive.
- **Team Development and Leadership:** Building team capability, resolving conflicts, motivating team members, and developing a collaborative, high-performing environment.
- **Stakeholder Engagement Execution:** Actively working to maintain stakeholder involvement, manage expectations, and convert stakeholders from resistant to supportive.
- **Procurement Execution:** Conducting procurements: soliciting bids, selecting sellers, and managing contract execution during the project.
- **Issue Log:** A document tracking issues (problems currently occurring), their owner, priority, and resolution status. Distinct from the risk register (future events).
- **Change Requests:** Formal requests to modify baselines, documents, or plans. All approved changes must be implemented during execution and documented.
- **Lessons Learned (Execution):** Capturing insights during execution about what is working and what is not, to enable real-time improvement and post-project organizational learning.
- **Information Distribution:** Ensuring project information reaches stakeholders through appropriate channels with the right frequency and format as defined in the communications plan.
- **Agile Execution Ceremonies:** Daily standup (sync on progress/blockers), sprint review (demo to stakeholders), sprint retrospective (team improvement) — core execution events in adaptive projects.

Key Terms & Definitions

Term	Definition
Deliverable	Any unique, verifiable product, result, or capability that is produced during a project phase or at project completion.
Quality Assurance	The process of auditing quality requirements and results from quality control measurements to ensure standards and definitions are being used correctly.
Issue Log	A document recording active problems or concerns that require management attention and resolution during project execution.
Change Request	A formal proposal to modify a project document, deliverable, or baseline. Must be reviewed and approved through change control.
Daily Standup	A short daily meeting (15 minutes max) in agile projects where team members report what they completed, what they plan to do, and any blockers.
Sprint Review	A meeting at the end of each sprint where the team demonstrates completed work to stakeholders for feedback and acceptance.
Sprint Retrospective	A meeting at the end of each sprint where the team reflects on its processes and identifies improvements for the next sprint.
Servant Leader	A leadership style where the project manager facilitates, coaches, and removes impediments rather than directing work — fundamental to agile project execution.
Procurement	The process of obtaining goods and services from external sources through purchasing, contracts, or service agreements.

Exam Tips

✦ EXAM TIPS

- Quality Assurance (QA) is execution-phase; Quality Control (QC) is monitoring and controlling phase. QA checks processes; QC checks deliverables. Get these straight!
- The issue log is used for current problems; the risk register is for potential future events. When something that was a risk becomes a reality, it becomes an issue.
- In agile execution, the Scrum Master is the facilitator and impediment remover — NOT the project manager directing the team. This servant leadership distinction is heavily tested.
- All changes, even verbally approved by the sponsor, must go through formal change control in predictive projects. Oral approvals are not sufficient.
- Daily standups are for the team — not status meetings for the project manager. The PM facilitates; the team reports to each other.

• During execution, if actual performance deviates from the plan, the PM should first determine the root cause, then take corrective action — not immediately replan or escalate.

Knowledge Check

✦ KNOWLEDGE CHECK

1. A team member discovers a defect in a deliverable that has already been accepted by the client. What should the project manager do?

2. A vendor is consistently delivering late, threatening the project schedule. The project manager has already communicated the issue twice. What are the next appropriate steps?

3. In an agile project, a team member asks the project manager to assign them specific tasks each morning. How should the project manager respond, and why?

4. During execution, the sponsor verbally approves a scope expansion. How should the project manager handle this, and what is the risk of proceeding without formal documentation?

5. A key stakeholder who was initially resistant to the project has become an active blocker. What execution-phase tools or strategies would you use to address this?

Lecture 8: Focus Area 4 — Monitoring and Controlling

Overview

The Monitoring and Controlling focus area encompasses all activities related to tracking, reviewing, and regulating project progress and performance against the project plan. It is the discipline that allows project managers to detect problems early, identify variances from baselines, and take corrective or preventive action before issues escalate into crises. Unlike the other focus areas, monitoring and controlling runs in parallel with all other project activities — it is continuous, not sequential.

The core of monitoring and controlling is measurement: comparing actual performance (actual cost, schedule performance, scope delivered) against the approved baselines and interpreting the variances. Earned Value Management (EVM) is the primary quantitative tool for this analysis, providing objective measures of schedule and cost performance. Beyond EVM, monitoring and controlling includes change control, risk monitoring, quality control, and stakeholder communication.

On the PMP exam, monitoring and controlling questions are among the most analytical. Expect calculations involving EVM metrics (CPI, SPI, EAC, ETC), interpretation of variance reports, and scenario-based questions about when to escalate, when to rebaseline, and when to recommend corrective action versus preventive action.

Key Concepts

- **Earned Value Management (EVM):** An integrated approach to tracking project performance by measuring actual work completed (Earned Value) against planned work and actual costs.
- **Key EVM Metrics:** PV (Planned Value), EV (Earned Value), AC (Actual Cost), SV (Schedule Variance = EV-PV), CV (Cost Variance = EV-AC), SPI (EV/PV), CPI (EV/AC).
- **Forecast Metrics:** EAC (Estimate at Completion) = BAC/CPI; ETC (Estimate to Complete) = EAC - AC; TCPI (To-Complete Performance Index) = (BAC-EV)/(BAC-AC).
- **Variance Analysis:** The process of comparing planned vs. actual results to determine the root cause of deviations and the corrective action required.
- **Change Control Process:** The formal process for submitting, reviewing, approving/rejecting, and implementing changes to project baselines. Outputs: approved change requests, updates to plans.
- **Quality Control (QC):** Inspecting deliverables to verify they meet quality standards. Tools: checklists, inspections, statistical sampling, control charts.
- **Risk Monitoring:** Continuously tracking identified risks, monitoring residual risks, identifying new risks, and evaluating risk response effectiveness.
- **Trend Analysis:** Examining project performance data over time to identify patterns that may indicate future problems before they occur.
- **Corrective Action vs. Preventive Action:** Corrective action addresses current problems; preventive action addresses potential future problems. Both may trigger change requests.
- **Rebaselining:** Updating an approved baseline to reflect approved changes. Only done after formal change control approval — never done unilaterally.

Key Terms & Definitions

Term	Definition
Planned Value (PV)	The authorized budget assigned to scheduled work — what you planned to accomplish by a given date.
Earned Value (EV)	The measure of work performed expressed in terms of the budget authorized for that work — what you have actually accomplished.
Actual Cost (AC)	The realized cost incurred for work performed during a specific time period.
Cost Performance Index (CPI)	CPI = EV/AC. Values > 1 are favorable (under budget); values < 1 are unfavorable (over budget).
Schedule Performance Index (SPI)	SPI = EV/PV. Values > 1 are favorable (ahead of schedule); values < 1 are unfavorable (behind schedule).
Estimate at Completion (EAC)	The expected total cost of the project at completion. Common formula: EAC = BAC/CPI.
Change Control Board (CCB)	The group authorized to review and approve/reject change requests to project baselines.
Control Chart	A graph used to determine whether project processes are in statistical control; tracks performance over time relative to upper and lower control limits.
Variance at Completion (VAC)	VAC = BAC - EAC. Indicates the projected surplus (positive) or deficit (negative) at project completion.

Exam Tips

✦ EXAM TIPS

- Memorize all EVM formulas: CPI = EV/AC, SPI = EV/PV, EAC = BAC/CPI, ETC = EAC - AC, VAC = BAC - EAC, TCPI = (BAC-EV)/(BAC-AC). Calculations will appear on the exam.
- CPI < 1 = over budget; SPI < 1 = behind schedule. Anything less than 1 is bad for both indices.
- Corrective action fixes what's already wrong; preventive action avoids future problems. Both require change requests in predictive projects.
- The exam frequently tests rebaselining scenarios. Rebaselining requires formal change control approval — a sponsor cannot verbally authorize it.
- In adaptive projects, monitoring is done through burndown charts, velocity tracking, and sprint reviews — not EVM. Know the adaptive equivalents.
- When a control chart shows data points outside control limits, it signals a process problem requiring investigation — not just an observation.

Knowledge Check

✦ KNOWLEDGE CHECK
1. A project has BAC = $500,000, PV = $200,000, EV = $180,000, AC = $210,000. Calculate CPI, SPI, EAC, and VAC. What does this data tell you about the project's health?
2. The project manager notices a consistent downward trend in SPI over the past 4 weeks, despite assurances from team leads that 'everything is on track.' What should the PM do?
3. A sponsor asks you to update the cost baseline to match current actual spending. What should you tell the sponsor, and what process is required?
4. During a quality control inspection, 8 defects are found in a batch of 50 units. How do you determine whether this is within acceptable limits, and what tools would you use?
5. An agile project team uses a burndown chart to track sprint progress. With 5 days remaining, the burndown line is significantly above the ideal line. What does this indicate, and what are your options?

Lecture 9: Focus Area 5 — Closing

Overview

The Closing focus area covers all activities required to formally complete or terminate a project, phase, or contract. Closing is often the most neglected focus area in practice — teams move on to the next project before completing the formal close of the current one. Yet PMBOK Guide 8th Edition emphasizes that proper closing is essential for benefits realization, organizational learning, stakeholder satisfaction, and resource release.

Closing activities include: obtaining formal acceptance of deliverables from the customer or sponsor, transitioning the product to operations or the customer, releasing project resources back to the organization, archiving project documents, closing procurement contracts, and conducting a final lessons learned session. In adaptive projects, closing activities occur at the end of each release as well as at the project's conclusion.

On the PMP exam, closing is tested both as a knowledge area and as a situational judgment topic. Candidates must know when a project should be closed (including early termination), what documents are produced during closing, and how closing connects to organizational process assets and benefits realization.

Key Concepts

- **Formal Acceptance:** The process of receiving formal sign-off from the customer or sponsor that deliverables meet requirements. Essential output of project closing.
- **Administrative Closure:** The process of finalizing all project activities — archiving documents, releasing resources, closing procurement contracts, and updating organizational assets.
- **Lessons Learned Repository:** The organizational database where project lessons learned are stored. A critical organizational process asset updated during closing.
- **Resource Release:** Formally releasing project team members back to their functional managers or resource pools upon project completion.
- **Procurement Closure:** Completing all open contracts, processing final invoices, and obtaining confirmation that all contract requirements have been met.
- **Benefits Realization Confirmation:** During closing, the PM verifies that the project has delivered the expected business benefits or confirms that a benefits realization plan is in place for post-project monitoring.
- **Product Transition:** The formal handover of the project's product, service, or result to the operations team or customer for ongoing use and support.
- **Final Performance Reporting:** Preparing the final project performance report summarizing scope, schedule, cost, quality, and stakeholder satisfaction outcomes.
- **Early Project Termination:** When a project is cancelled before completion, closing activities are still required — partial deliverables must be documented, resources released, and lessons learned captured.
- **Project Archives:** The complete documentation of the project — plans, changes, reports, communications, and contracts — stored for future reference and organizational learning.

Key Terms & Definitions

Term	Definition
Formal Acceptance	Written acknowledgment from the customer or sponsor that project deliverables meet the agreed requirements.
Lessons Learned	Documented knowledge gained during a project that can be applied to future projects to improve performance.
Administrative Closure	The procedural work of officially completing the project: archiving records, releasing resources, and updating organizational assets.
Procurement Closure	The formal close of all contracts and procurement agreements, including final payments and performance evaluations of vendors.
Organizational Process Assets (OPA)	Knowledge repositories, policies, procedures, and historical data from past projects that organizations use to improve future projects.
Benefits Realization Plan	A document describing how and when project benefits will be measured and realized, often managed post-project by the business.
Project Closure Report	A document summarizing project performance, lessons learned, and the final status of all deliverables at project close.
Product Transition	The planned transfer of project deliverables to the operations team, customer, or ongoing support function.
Final Lessons Learned Session	A structured meeting held at project close to capture what worked, what didn't, and recommendations for future projects.

Exam Tips

✦ EXAM TIPS

- Lessons learned are captured throughout the project but are FORMALLY documented and stored during project closing. The exam will ask about both timing aspects.
- Even cancelled (terminated early) projects require formal closing activities. Never skip close-out because 'the project was cancelled' — the exam penalizes this.
- Formal acceptance must be in writing. A verbal 'looks good' from the customer is NOT formal acceptance. Always get written sign-off.
- The difference between administrative closure and procurement closure: admin closure closes the project; procurement closure closes vendor agreements. Both must happen.
- The PM is responsible for closing — not the sponsor. If the sponsor wants to skip closing to 'move on,' the PM must advocate for proper close-out.
- Benefits realization often continues AFTER the project closes. The PM hands off to the business owner who monitors ongoing benefits — the exam tests this post-project responsibility.

Knowledge Check

✦ KNOWLEDGE CHECK

1. A project sponsor declares the project complete and immediately assigns team members to a new project without conducting formal close-out. What risks does this create?

2. A long-term project is terminated early due to budget cuts. What closing activities must still be performed, and why are they important?

3. A customer verbally accepts the final deliverable but refuses to sign the formal acceptance document. What should the project manager do?

4. At project close, the PM realizes that benefits realization will take 18 months post-project to fully manifest. Who is responsible for monitoring these benefits, and what document guides this?

5. The project archives must be stored for 7 years per organizational policy. What types of documents should be included, and who is responsible for ensuring they are complete?

Lecture 10: Focus Areas as a Whole — Interactions in Predictive vs Adaptive Projects

Overview

Having studied each focus area individually, it is essential to understand how they interact as a system and how those interactions differ between predictive and adaptive projects. In PMBOK Guide 8th Edition, focus areas are not sequential stages — they overlap, iterate, and inform each other continuously. The project manager's ability to navigate these interactions is what separates reactive management from proactive leadership.

In a predictive project, the focus areas flow in a roughly sequential pattern: initiating establishes the project, planning defines the approach, executing performs the work, monitoring and controlling keeps work on track, and closing completes the project. However, even in predictive projects, monitoring runs concurrently with executing, and planning may need to be revisited when significant changes occur.

In an adaptive project, the focus areas recur in every sprint or iteration — mini versions of initiating (sprint planning), executing (sprint development), and closing (sprint review and retrospective) happen every 1-4 weeks. Monitoring occurs daily via standups and burndown charts. This recursive pattern creates a continuous learning and improvement cycle that is the defining characteristic of agile project management.

Key Concepts

- **Focus Area Integration:** No focus area operates in isolation. Planning outputs feed execution; execution produces data for monitoring; monitoring triggers replanning or correction; closing feeds lessons learned into future initiating.
- **Concurrency of Focus Areas:** Monitoring and controlling always runs in parallel with execution. Planning may recur when significant changes are approved. Closing begins when deliverables are complete.
- **Predictive Flow Pattern:** Initiating to Planning to Executing with Monitoring and Controlling running in parallel, then Closing. Formal handoffs between each stage with documented outputs.
- **Adaptive Recursive Pattern:** Each sprint is a miniature project cycle: Sprint Planning (initiating+planning), Sprint Execution, Sprint Review (demo/closing), Retrospective (improvement), then the next sprint begins.
- **Tailoring Focus Area Application:** In hybrid projects, different components may be in different focus areas simultaneously. A PM must manage multiple cadences and track where each component is in its lifecycle.
- **Feedback as a Control Mechanism:** Both approaches use feedback loops — predictive uses formal change control; adaptive uses sprint reviews. Feedback from monitoring informs replanning and execution adjustment.
- **Progressive Elaboration:** The ongoing refinement of project details as more information becomes available. Applies to both predictive (rolling wave) and adaptive (backlog refinement) projects.
- **Knowledge Flow Across Focus Areas:** Lessons from execution flow to monitoring; monitoring findings trigger planning updates; planning changes are executed; execution outputs are closed. This knowledge cycle is continuous.

- **Governance Checkpoints:** Predictive: formal phase gates. Adaptive: sprint reviews and release demos. Hybrid: a mix of both. Governance ensures value delivery at each checkpoint.
- **Stakeholder Engagement Across Focus Areas:** Stakeholders are most active during initiating and closing in predictive projects. In adaptive projects, stakeholders engage at every sprint review — maintaining continuous alignment.

Key Terms & Definitions

Term	Definition
Focus Area	A grouping of related project management activities that describes WHAT needs to be done at different points in the project (PMBOK 8 replaces Process Groups with Focus Areas).
Process Group (Legacy)	The traditional PMBOK categorization: Initiating, Planning, Executing, Monitoring & Controlling, Closing. Focus Areas cover the same ground in PMBOK 8.
Sprint Cycle	The recurring mini-project loop in adaptive projects comprising sprint planning, execution, review, and retrospective.
Progressive Elaboration	The iterative process of increasing the level of detail in project plans as more information and greater clarity becomes available.
Feedback Loop	A structured mechanism for using output information to influence future inputs — the core of continuous improvement in both predictive and adaptive approaches.
Governance Checkpoint	A structured review point (phase gate or sprint review) where performance is assessed and authorization to continue is granted.
Change Management	The process of managing approved changes to project baselines through formal control procedures, ensuring traceability and alignment.
Benefits Realization Cycle	The end-to-end flow from project initiating through execution, delivery, and post-project monitoring to confirm that intended business value has been achieved.

Exam Tips

✦ EXAM TIPS

- The PMP exam frequently presents scenarios where a PM tries to apply purely predictive or purely adaptive thinking to a situation that requires the other. Always assess the context.
- Know that monitoring and controlling is ALWAYS concurrent with execution — it is never finished before execution begins or completed after execution ends.

- Sprint retrospectives are the adaptive equivalent of lessons learned. The key difference: retrospectives happen after every sprint; lessons learned are captured at project close in predictive.
- In hybrid projects, you must manage multiple cadences simultaneously. Tracking which components are in which focus area at any given time requires discipline and clear communication.
- The PMP exam is application-based — it tests whether you know WHEN to apply each focus area's tools and WHAT to do when focus areas overlap or conflict.
- Benefits realization links closing back to initiating — the business case from initiating defines the expected benefits that closing (and post-project) confirms were achieved.

Knowledge Check

✦ KNOWLEDGE CHECK

1. A predictive project is in execution when a major scope change is requested. Which focus areas are activated by this change request, and in what order?
2. In an agile project, a stakeholder claims that 'we never do planning.' How would you explain the planning activities that occur in each sprint?
3. A hybrid project has software (adaptive) and hardware (predictive) components. The hardware is in planning while software is in its 4th sprint. How do you manage governance for both simultaneously?
4. A project manager says 'I will capture lessons learned at the end of the project.' An agile coach says lessons should be captured every sprint. Who is right, and why?
5. How does the benefits realization cycle connect the closing focus area back to the initiating focus area? What document from initiating should be revisited during closing?

Lecture 11: Section 3 Summary and Knowledge Check

Overview

This closing lecture consolidates Section 3 into one integrated picture: project phases and phase gates provide the time-bound structure every project moves through; development approach (predictive, adaptive, or hybrid) is the overall strategy for creating deliverables, selected from deliverable, project, and organizational factors; delivery cadence (single, multiple, or periodic) determines how often deliverables reach stakeholders; and the five focus areas -- Initiating, Planning, Executing, Monitoring and Controlling, and Closing -- apply to every project, sequentially in predictive projects and cyclically within each iteration in adaptive projects.

The lecture also previews the exam's favorite scenario types for this section: recommending a development approach from a set of project factors, identifying what belongs in a project charter, matching activities to the correct focus area, handling a change request through change control, and contrasting sequential versus cyclical focus-area behavior.

The core exam-ready takeaway is that PMBOK 8 expects tailoring, not a single correct method -- the right answer to a scenario is the approach and cadence that best fit the deliverable, the project, and the organization described.

Key Concepts

- Project phases divide the project into distinct, time-bound segments; phase gates are governance checkpoints between phases where the team decides whether to proceed.
- Development approach selection depends on three factor categories: deliverable/product factors (requirement stability, complexity, criticality), project factors (duration, team size/experience, stakeholder availability), and organizational factors (maturity, standards, industry norms).
- Stable requirements and lower complexity point toward predictive; uncertain or changing requirements and higher complexity point toward adaptive; most real-world projects sit somewhere on the spectrum as hybrids.
- Delivery cadence -- single, multiple, or periodic delivery -- determines how frequently stakeholders receive deliverables, independent of the development approach chosen.
- The five focus areas (Initiating, Planning, Executing, Monitoring and Controlling, Closing) apply to every project regardless of approach.
- In predictive projects, focus areas are roughly sequential: Initiating -> Planning -> Executing (with Monitoring and Controlling in parallel) -> Closing, with approved changes triggering replanning.
- In adaptive projects, all five focus areas cycle within every iteration: a short Initiating/Planning/Executing/Monitoring and Controlling/Closing loop repeats, with just-in-time planning replacing upfront planning.
- The project charter is the key output of Initiating: objectives, scope, success criteria, stakeholders, constraints, assumptions, and formal authorization.
- Change requests are handled through change control: impacts are evaluated and a recommendation to approve or reject follows.
- Tailoring is the unifying theme of Section 3 -- there is no single right way to run a project; the PM adapts phases, approach, cadence, and focus-area rhythm to context.

Key Terms & Definitions

Term	Definition

Phase gate	A governance checkpoint between project phases where stakeholders decide whether to continue, change direction, or stop the project.
Just-in-time planning	Planning only the current iteration in detail rather than the whole project upfront; characteristic of adaptive focus-area cycling.
Development approach factors	The three categories -- deliverable/product, project, and organizational -- used to recommend predictive, adaptive, or hybrid on the exam.
Sequential focus areas	The predictive-project pattern where Initiating, Planning, Executing, Monitoring and Controlling, and Closing occur largely in order.
Cyclical focus areas	The adaptive-project pattern where all five focus areas repeat within every iteration.

Exam Tips

✦ EXAM TIPS

1. When a scenario lists project factors and asks for a recommended approach, evaluate requirement stability and complexity first -- that pairing resolves most predictive-vs-adaptive questions.
2. Memorize the project charter's contents (objectives, scope, success criteria, stakeholders, constraints, assumptions, authorization) -- it is a frequently tested Initiating output.
3. If a scenario shows focus areas repeating rapidly with short planning windows, that signals adaptive cycling, not a predictive sequence.
4. Every change request answer should route through change control -- evaluate impact, then approve or reject; skipping straight to implementation is a common wrong answer.

Knowledge Check

✦ KNOWLEDGE CHECK

1. What three categories of factors guide development-approach selection, and how does each push toward predictive versus adaptive?
2. How do the five focus areas behave differently in a predictive project compared to an adaptive project?
3. What does the project charter contain, and which focus area produces it?
4. Why does PMBOK 8 treat tailoring as the unifying theme across phases, approach, cadence, and focus areas?

Appendix: Quick Reference

Development Approach Comparison

Factor	Predictive	Adaptive (Agile)	Hybrid
Requirements	Fixed upfront	Evolve via backlog	Mixed (stable + evolving)

Planning	Front-loaded, comprehensive	Rolling, per sprint	Both levels simultaneously
Deliverables	End of project (single)	Each sprint/release (periodic)	Batched or periodic
Change	Formal CCB required	Welcomed via backlog	Formal for stable; agile for evolving
Risk Level	Lower uncertainty	Higher uncertainty	Mixed risk profile
Stakeholder Engagement	Phase gates	Every sprint review	Governance + sprint reviews
Best For	Construction, compliance	Software, innovation, R&D	Complex, multi-component
PM Role	Plan, direct, control	Facilitate, coach, remove impediments	Context-dependent

Development Approach Selection Factors

Selection Factor	Approach Guidance
Requirements stability	Stable → Predictive | Unstable/evolving → Adaptive
Uncertainty & risk level	Low uncertainty → Predictive | High uncertainty → Adaptive
Regulatory compliance needs	High regulation → Predictive | Low/no regulation → Adaptive
Customer engagement availability	Available for frequent reviews → Adaptive | Periodic reviews only → Predictive
Team agile maturity	High maturity → Adaptive | Low maturity → Predictive or Hybrid
Delivery timing requirements	Value ASAP → Adaptive | All at once acceptable → Predictive
Product novelty	High novelty/innovation → Adaptive | Proven/standard → Predictive
Team size & distribution	Small, co-located → Adaptive | Large, distributed → Predictive or Hybrid
Organizational culture	Collaborative, empowering → Adaptive | Hierarchical, process-driven → Predictive

EVM Formula Cheat Sheet

Formula	Description	Interpretation

SV = EV - PV	Schedule Variance	Positive = ahead; Negative = behind schedule
CV = EV - AC	Cost Variance	Positive = under budget; Negative = over budget
SPI = EV / PV	Schedule Performance Index	> 1 = ahead of schedule; < 1 = behind
CPI = EV / AC	Cost Performance Index	> 1 = under budget; < 1 = over budget
EAC = BAC / CPI	Estimate at Completion	Projected total cost at current performance
ETC = EAC - AC	Estimate to Complete	Remaining cost to finish the project
VAC = BAC - EAC	Variance at Completion	Projected surplus (+) or deficit (-) at end
TCPI = (BAC-EV)/(BAC-AC)	To-Complete Performance Index	Efficiency needed to meet BAC; > 1 = harder than current

Key Terms Master List

Term	Definition
Adaptive Approach	Agile/iterative development that delivers incremental value and welcomes change.
Backlog	Ordered list of all project work, prioritized by the Product Owner.
Business Case	Document justifying the project investment with expected benefits.
Change Control Board (CCB)	Group authorized to review/approve changes to project baselines.
CPI	Cost Performance Index = EV/AC; measures cost efficiency.
Delivery Cadence	The timing and frequency of project deliverable releases.
EVM	Earned Value Management - integrates scope, schedule, and cost performance measurement.
Focus Area	PMBOK 8 grouping of related project activities describing what needs to be done.
Formal Acceptance	Written customer/sponsor sign-off that deliverables meet requirements.
Hybrid Approach	A tailored blend of predictive and adaptive practices.
Iteration / Sprint	Fixed-length development cycle (1-4 weeks) in adaptive projects.
Lessons Learned	Documented knowledge from a project stored to improve future projects.

Phase Gate	Go/No-Go review at the end of a project phase.
Predictive Approach	Waterfall/plan-driven development with fully defined upfront scope.
Project Charter	Document authorizing the project and granting PM authority.
Project Life Cycle	Full series of project phases from initiation to closure.
Rolling Wave Planning	Detailed near-term planning with high-level planning for future work.
SPI	Schedule Performance Index = EV/PV; measures schedule efficiency.
Sprint Retrospective	Agile ceremony where team reflects on process improvements after each sprint.
Sprint Review	Agile ceremony where team demos work to stakeholders for feedback.
Tailoring	Intentional adaptation of project methods to the specific project context.
WBS	Work Breakdown Structure - hierarchical decomposition of all project deliverables.